Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Ray, Chester A. Ph.D.		
Project Title:	Ultrasonic Bone Stimulation: Countermeasure to Orthostatic Intolerance		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Hershey	**State:**	PA
Zip Code:	17033-2390	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2004	**End Date:**	05/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-CA00404		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Orthostatic intolerance (OI) is a major physiological consequence of spaceflight. It has been estimated that nearly one-half to two-thirds of all astronauts experience OI after spaceflight. Post-spaceflight OI is a serious medical issue for manned-spaceflight because physical performance of an astronaut is severely impaired at a time when rapid egress from the spacecraft may be needed during an emergency. A number of physiological factors have been suggested to contribute to this problem with a number of countermeasures developed to prevent or alleviate post-spaceflight OI. However, as of yet, no single countermeasure has effectively prevented post-spaceflight OI. The overall objective of this project is to determine whether ultrasonic bone stimulation of the mastoid improves orthostatic tolerance and thus could serve as an effective and simple countermeasure for post-spaceflight OI. Based on preliminary data, it is hypothesized that ultrasonic bone stimulation of the mastoid will improve orthostatic tolerance in OI patients and in subjects exposed to bed rest. A secondary goal of this project is to determine the mechanism by which ultrasonic bone stimulation of the		

	mastoid increases orthostatic tolerance. It is hypothesized that this novel countermeasure acts to increase sympathetic nerve activity through engagement of the vestibulosympathetic reflex. Direct measurement of muscle sympathetic nerve activity will be made during ultrasonic bone stimulation in young, aged, and vestibular deficient subjects. This project should provide clear information on the possible role of ultrasonic bone stimulation of the mastoid as an effective and simple countermeasure for OI following spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current project may assist in our understanding of the mechanisms responsible for orthostatic intolerance and/or orthostatic hypotension in humans on earth. Additionally, the device being tested to elicit ultrasonic bone stimulation may be develop to help individuals on earth who experience orthostatic intolerance. This device is noninvasive and pose miminal risk to the patient.
Task Progress:	The overall objective of this project is to determine autonomic mechanisms related to orthostatic intolerance following space flight and to start the development of a simple and effective countermeasure for this problem. The specific aims and results of each project conducted this past year is stated below. Specific Aim 1: To study the effects of prolonged bed rest on the vestibulosympathetic reflex (VSR). Result: We tested the hypothesis that the VSR would be attenuated after approximately 35 days of head-down bed rest (HDBR). MSNA and leg blood flow were measured before and after 36±1 days of HDBR in 7 subjects. The results from this study demonstrates a significant attenuation of the VSR after prolonged HDBR. These data provide support for the concept that attenuation of the VSR could contribute to an inadequate increase in MSNA observed post-HDBR in those subjects that experience orthostatic intolerance. Specific Aim 2: To determine the effect of ultrasonic mastoid stimulation on orthostatic tolerance following one-day HDBR. Result: We conducted head-up tilt test on 25 subjects after one-day HDBR with and without ultrasonic bone stimulation (UBS) (HiSonic®). Twelve out of the 25 subjects showed an increase in tilt time using UBS. The limitation of this study was that tilt was limited to 30 minutes and that one-day bed rest did not induce presyncopal symptoms in all subjects. These initial studies do show that UBS may improve orthostatic intolerance following bed rest. Specific Aim 3: To determine the effect of ultrasonic mastoid stimulation on MSNA. Results: To test the effect of artificial activation of the otolith organs, UBS was applied to the mastoid bone and the clavicle and compared to head down rotation. Our results from this study demonstrate that UBS can produce increased sympathetic outflow similar to that seen during natural stimulation. Specific Aim 4: To test the hypothesis that the VSR modulation of MSNA would be unaffected by baroreceptor unloading and would be inhibited during baroreceptor loading. Results: The findings from our study supported our hypothesis. Specific Aim 5: To test the influence of dramamine (motion sickness drug) on sympathetic neural responses to orthostatic stress and vestibular activation in humans. Result: Dramamine augments the cardiac-vagal baroreflex, but does not alter resting MSNA, the sympathetic baroreflex, or the VSR. Specific Aim 6: To test the hypothesis that endurance training reduces renal vasoconstriction to orthostatic stress. Result: The findings from our study supported our hypothesis. Specific Aim 7: To test the hemodynamic effects of mental stress during orthostatic stress. Result: Augmented sympathetic activity of the upright posture does not alter vascular responses of the visceral organs or forearm to MS. Specific Aim 8: To test the effect of aldosterone on the baroreflex in humans. Result: Aldosterone impairs the arterial baroreflexes in humans.
Bibliography Type:	Description: (Last Updated: 01/26/2007)
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